INCH-POUND
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DETAIL SPECIFICATION

COOLERS, FLUID, INDUSTRIAL, AIR, MOTOR AND GENERATOR, NAVAL SHIPBOARD

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers fluid industrial air coolers for motors, generators, and similar electrical equipment for naval shipboard application. These coolers commonly form part of the water-air-cooled enclosures built in accordance with MIL-DTL-2036.

1.2 Classification. The coolers are of the following classes as specified (see 6.2):

a. Class 1 - Submarine, salt water cooled, salt water side subject to sea pressure for greater than 200 feet submergence, cyclic life stipulated.

b. Class 2 - Submarine, salt water cooled, salt water side secured for greater than 200 feet submergence.

c. Class 3 - Surface ship, salt water cooled.

d. Class 4 - Submarine and surface ship fresh water cooled.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-B-639 - Brass, Naval: Flat Products

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections

QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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TT-C-490 - Chemical Conversion Coatings and Pretreatments for Metallic Substrates (Base for Organic Coatings)

TT-P-1757 - Primer Coating, ALKYD Base, One Component

FEDERAL STANDARDS

FED-STD-H28 - Screw-Thread Standards for Federal Services

COMMERCIAL ITEM DESCRIPTIONS

A-A-59004 - Anti-Galling Compound, Thread Lubricating, Seizing Resistant, and Calcium Hydroxide Containing

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impacts); Shipboard Machinery, Equipment and Systems, Requirements for

MIL-PRF-907 - Antiseize Thread Compound, High Temperature

MIL-PRF-1149 - Gasket Materials, Synthetic Rubber, 50 and 65 Durometer Hardness

MIL-DTL-1222 - Studs, Bolts, Screws and Nuts for Applications where a High Degree of Reliability is Required; General Specification for

MIL-PRF-6855 - Rubber, Synthetic, Sheets, Strips, Molded or Extruded Shapes, General Specification for

MIL-T-15005 - Tubes, Condenser and Heat Exchanger, Copper Nickel Alloys (UNS C70600 &C71500)

MIL-DTL-15024 - Plates and Tags for Identification of Equipment, General Specification for

MIL-P-15024/5 - Plate, Identification

MIL-DTL-15090 - Enamel, Equipment, Light-Gray (Formula No. 111)

MIL-C-15726 - Copper-Nickel Alloy, Sheet, Plate, Strip, Bar, Rod and Wire

MIL-E-15809 - Expander, Tube, Condenser and Heat Exchangers

MIL-T-16420 - Tube, Copper-Nickel Alloy, Seamless and Welded (Copper Alloy Numbers 715 and 706)

MIL-DTL-18001 - Anodes, Sacrificial Zinc Alloy

MS18229 - Plug for O-Ring Gasket

MIL-A-19521 - Anode Retaining Support Plugs and Anode Selection and Installation Design Criteria for Shipboard Condensers and Heat Exchangers

MIL-G-21610 - Gaskets, Heat Exchanger, Various Cross Section Rings, Synthetic Rubber

MIL-S-22698 - Steel Plate, Shapes and Bars, Weldable Ordinary Strength and Higher Strength: Structural

MIL-N-24106 - Nickel-Copper Alloy Bars, Rods, and Forgings

MIL-T-24107 - Tube, Copper (Seamless) (Copper Alloy Numbers C10100, C10200, C10300, C10800, C12000, C12200, and C14200)

MIL-L-24131 - Lubricant, Colloidal Graphite in Isopropanol

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MIL-B-24480 - Bronze, Nickel-Aluminum (UNS No. C95800) Castings for Seawater Service

MIL-PRF-24508 - Grease, High Performance, Multi-Purpose

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-22 - Welded Joint Designs

MIL-STD-777 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships

DOD-STD-1399-301 - Interface Standard for Shipboard Systems, Ship Section 301 Motion and Attitude (Metric)

MIL-STD-2193 - Ship Hydraulic System Components

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment

MIL-HDBK-470 - Designing and Developing Maintainable Products and Systems, Volume 1

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. GENERAL SERVICES ADMINISTRATION

GSA Global Supply Catalog

(Copies of this document are available online at https://www.gsaglobalsupply.gsa.gov.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-5959186 - Submarine Heat Exchanger Anode Plug

810-1385861 - Flanges Seawater 700 PSI Max

(Copies of these documents are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of this document may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVSEA PUBLICATIONS

0900-LP-001-7000 - Fabrication and Inspection of Brazed Piping Systems

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of these documents are available online via TDMIS at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS 411-1 - Hazardous Material Target List

(Copies of this document are available online at www.aia-aerospace.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.12 - Class 5 Interference-Fit Thread

ASME BPVC Section III - Rules for Construction of Nuclear Facility Components

ASME BPVC Section VIII, Division I - Rules for Construction of Pressure Vessels Division I

(Copies of these documents are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A106/A106M - Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service

ASTM B98/B98M - Standard Specification for Copper-Silicon Alloy Rod, Bar and Shapes

ASTM B139/B139M - Standard Specification for Phosphor Bronze Rod, Bar, and Shapes

ASTM B171/B171M - Standard Specification for Copper-Alloy Plate and Sheet for Pressure Vessels, Condensers, and Heat Exchangers

ASTM B265 - Standard Specification for Titanium and Titanium Alloy Strip, Sheet, and Plate

ASTM B338 - Standard Specification for Seamless and Welded Titanium and Titanium Alloy Tubes for Condensers and Heat Exchangers

ASTM B369/B369M - Standard Specification for Copper-Nickel Alloy Castings

ASTM B381 - Standard Specification for Titanium and Titanium Alloy Forgings

ASTM D5363 - Standard Specification for Anaerobic Single-Component Adhesives (AN)

(Copies of these documents are available online at www.astm.org.)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS

IEEE/ASTM SI 10 - American National Standard for Metric Practice

(Copies of this document are available online at www.ieee.org.)

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SAE INTERNATIONAL

SAE AMS7259	-	Rubber: Fluorocarbon (FKM) High Temperature/Fluid Resistant Low Compression Set/85 to 95 Hardness For Seals In Fuel Systems and Specific Engine Oil Systems
SAE AMS8660	-	Silicone Compound NATO Code Number S-736
SAE AS568	-	Aerospace Size Standard for O-Rings
SAE J1926/1	-	Connections for General Use and Fluid Power - Ports and Stud Ends with ASME B1.1 Threads and O-Ring Sealing - Part 1: Threaded Port with O-Ring Seal in Truncated Housing

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.2.

3.1.1 Structural requirements. The structural design calculations for class 1 coolers shall be in accordance with Section III (class I vessel) of ASME Boiler and Pressure Vessel Code (BPVC). Structural design calculations for coolers other than class 1 shall be in accordance with ASME BPVC Section VIII, Division I. The waterside design pressure shall be as specified (see 6.2) or shall be a minimum of 50 pounds per square inch (lb/in^2).

3.1.1.1 Metric conversion. When metric conversion is required for machining purposes, dimensional conversion shall be in accordance with IEEE/ASTM SI 10.

3.1.2 Space and weight. Space occupied and weight of coolers shall be held to a minimum consistent with meeting the requirements of this specification and stipulated performance requirements.

3.1.2.1 Submarine installations. The dimensions of air coolers to be installed in submarines shall be such that the coolers can be removed through a standard hatch, or in case there is no hatch in the compartment in which the unit is to be installed, such that the unit can be moved through bulkhead openings into a compartment provided with a hatch. Hatch and bulkhead opening dimensions shall be as specified (see 6.2).

3.1.3 Requirements due to ship attitude or motion. Unless otherwise specified (see 6.2), coolers shall meet the requirements of DOD-STD-1399-301 with regard to ship motion and attitude.

3.1.4 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

3.1.5 Prohibited materials. Coolers shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS 411-1 for all other programs.

3.1.6 Welding and allied processes. Fabrication welding and inspection shall be in accordance with S9074-AR-GIB-010/278, except as modified herein. For class 1 coolers, welding on parts subjected to sea pressure shall be in accordance with class A-F of S9074-AR-GIB-010/278, except as modified herein. Welded joints on the salt water side of class 1 coolers shall be radiographable, except for vent and drain nipples. Waterbox vent and drain nipple welded connections of class 1 coolers shall meet the 100 percent weld efficiency requirements for pressure vessels.

3.1.7 Casting inspection and repair. Casting inspection and repair shall be in accordance with S9074-AR-GIB-010/278.

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3.1.8 Brazing. Brazing shall be in accordance with 0900-LP-001-7000, except that requirements for use of preinserted brazing rings are applicable only to pipe fittings. Brazed joints except where used in outer tube to inner tube sheet joints shall be ultrasonically inspected. Brazing shall not be used on the sea water side of submarine coolers (for example, class 1 or class 2 coolers).

3.1.9 Screw threads. Thread form, dimensions, and tolerances shall be in accordance with FED-STD-H28.

3.1.9.1 Threaded parts. Threaded parts shall be in accordance with to MIL-DTL-1222, except that materials shall be as specified in 3.2. The construction shall be such that standard GSA catalog wrenches can be used throughout. Tapered pipe threads shall not be used. The practice of bottoming or shouldering studs shall not be used. For the set end of studs, a class 3 fit used with locking compound in accordance with ASTM D5363, group 1, class 2, grade 3; or a class 5 interference fit in accordance with to ASME B1.12 shall be used. Threaded plugs, threaded connections, and threaded closures are prohibited in the submergence pressure boundary of class 1 coolers.

3.1.9.1.1 Thread lubricants. Threaded fasteners with torque requirements shall be lubricated before assembly. The lubricant used shall meet the following requirements as applicable:

a. For class 1 coolers, the lubricant for bolting subjected to submergence pressure shall be in accordance with A-A-59004.

b. For coolers in saturated steam plant ships, lubricant shall be graphite in isopropanol and shall be in accordance with MIL-L-24131.

c. For all other bolting, the lubricant shall be in accordance with MIL-PRF-907.

3.1.9.1.2 Preferred fastener types. Preferred fastener types are listed in order of preference:

a. Through bolt or through (two-nut) stud.

b. Tap-end stud (one-nut).

c. Cap screw or cap bolt.

3.1.9.1.3 Cap screws and cap bolts. Cap screws and cap bolts shall not be used for waterbox-to-shell or waterbox-to-tube sheet bolting, for inspection cover bolting, or for zinc anode support cover bolting.

3.1.9.1.4 Collar or stud bolts. Where collar bolts or stud bolts are used to make up a multiple flange joint, so that it can be selectively disassembled, such bolts or studs shall have a square extension beyond the threads on one end, for use of a wrench to prevent turning of the bolt when the nuts are tightened or removed.

3.1.9.1.5 Thread engagement. Stud engagement shall be sufficient to hold the maximum design load. In no case shall engagement be less than the nominal diameter of the fastener. Nuts shall be fully engaged in all cases. For a threaded fastener, not less than one thread but no more than four threads shall protrude beyond the crown of the nut.

3.1.9.1.6 Waterbox and tube sheet bolting. Waterbox and tube sheet bolting shall meet the following requirements:

a. The bolting shall be sequentially tightened to a stipulated torque value by means of a torque wrench.

b. For class 1 coolers, bolting construction including stipulated torque values shall be established based on supporting calculations.

c. For class 1 coolers, the bolting subjected to submergence pressure shall meet the following requirements:

(1) Calculated bolt stress due to design pressure shall be limited to ⅓ of the yield strength of the material at the design temperature.

(2) Maximum value of service stress averaged across bolt cross section and neglecting stress concentration factor shall be limited to ⅔ of the yield strength of the material of the design temperature.

(3) Maximum value of service stress at periphery of the bolt cross section resulting from tension, plus bending (neglecting stress concentration factor) shall be limited to the yield strength of the material at the design temperature.

(4) The value of installation torque specified on the drawings shall be calculated as follows:

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(a) Using P as the hydrostatic test load or load based upon design pressure plus maximum piping load on nozzles, whichever is greater, and high coefficient of friction (μ=0.13), determine torque using the following relation:

(P/T)μ = Load per unit torque corresponding to bolt size and lubricant

$$(P/T)\mu = \frac{12}{r_{t,}\left[\frac{\cos\theta\tan\alpha+\mu}{\cos\theta-\mu\tan\alpha}+\frac{r_c\mu}{r_t}\right]}$$

Where:

r_t = minimum pitch radius of external threads (inches)

ɵ = ½ of thread angle (degrees)

α = Helix angle (degrees) equal arc tan $[1/(2\pi r_t N)]$

r_c = Collar radius, average of $D_h/2$ and bolt major radius (inches)

N = Number of threads per inch

μ = 0.13

T = Torque (foot-pounds)

D_h = Diameter across flats of nuts (average of maximum and minimum) (inches)

D = Average of mean pitch and minor diameters of external thread (inches) equals $2r_t - (0.32476/N)$

(b) Using the installing torque above and low coefficient of friction (μ = 0.065), stress shall be determined from the following equation:

$$S_{av} = \frac{P}{6\pi\mu^2 D_h{}^2 D^3}\left[(D^2 + 16\mu^2 D_h{}^2)^{3/2} - D^3\right]$$

This calculated stress shall not exceed ⅔ of the yield strength of the bolt material. Refer to 3.3.6.2 for additional requirements on threaded fasteners securing waterbox inspection openings on class 1 coolers.

Note: P in equation above is preload force using the low coefficient of friction (μ = 0.065) and the installation torque calculated above.

(5) Fatigue analysis of bolts shall be in accordance with section III (class 1 vessel) of ASME BPVC.

(d) For class 1 coolers, all submergence pressure boundary studs and bolts shall be of a reduced shank construction to reduce bending stresses.

3.1.10 Shock resistance. All air coolers shall withstand the high-impact shock test as specified for grade A, class 1 equipment of MIL-DTL-901.

3.1.10.1 Coolers supported by the parent machine. Coolers that are to be supported by the parent machine shall withstand high-impact shock tests (see 4.5.3).

3.1.10.2 Coolers constructed to be separately mounted. Coolers constructed for separate mounting shall withstand high-impact shock tests (see 4.5.3).

3.1.10.3 Shock-resistant construction. Clearance for fasteners shall be no greater than the following, unless permitted by the applicable pipe flange standard:

Nominal bolt diameter (inch)	Nominal diameter of hole (inch)
¾ and smaller	Nominal bolt diameter plus 1/32
Larger than ¾	Nominal bolt diameter plus 1/16

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3.1.11 Water velocity and pressure drop limits. In the construction of cooling water spaces and connections, a smooth flow path shall be provided in order to minimize erosion-corrosion attack. Cooling water velocity at design point shall not exceed 7.5 feet per second at the flange face of the entrance nozzle or 6 feet per second through the tubes. For coolers using sea water as a coolant, the velocity through the tubes shall be not less than 4.5 feet per second. Unless otherwise specified (see 6.2), the maximum permissible pressure drop across the tube side shall be 6 lb/in².

3.1.12 Water inlet temperature. Unless otherwise specified for design purposes (see 6.2), temperature of coolant supply shall be taken as follows:

		Coolant	Temperature (°F)
a.	Submarines:	Sea water	85
		Fresh water	95
b.	Surface Ships:	Sea water	90
		Fresh water	95

3.1.13 Waterside pressures. The waterside design pressure shall be as specified (see 6.2) or shall be a minimum of 50 lb/in². The hydrostatic test pressure shall be 1½ times the design pressure.

3.1.14 Heat transfer surface. The amount of surface provided shall be based on the duty specified in British Thermal Units per hour (Btu/hr) (see 6.2), applying a 10 percent fouling factor to the heat transfer coefficient for clean tubes. Unless otherwise specified (see 6.2), while the machine is operating at rated electrical load, the air cooler shall cool the air so that the temperature of the air leaving the cooler shall not exceed 122 °F.

3.1.15 Inlet and outlet water connections. Inlet and outlet water connections shall be flanged, and the flanges shall be cast integrally or of the welding type. Unless otherwise specified (see 6.2), requirements for flanged connections on waterboxes shall be as follows:

a. Flanges shall be in accordance with the applicable schedule of MIL-STD-777 and 810-1385861 for surface ship and submarine air coolers, respectively, in regard to diameter, minimum thickness, drilling, and flange finish.

b. When flanges using O-ring gaskets are furnished, the groove for the O-ring shall be in the companion flange on the shipbuilder's piping.

3.1.16 Waterside vents and drains. Provision shall be made to permit complete venting and draining of the cooling water circuits. Waterside vent and drain connections shall be a minimum of ⅜-inch nominal pipe size (NPS) and shall be at least one NPS larger than the tell-tale system vent and drain connections (see 3.3.1.1). Such root connections shall have the wall thickness in accordance with MIL-T-16420, 70-30 copper nickel tube (class 6000 for 1 inch and smaller and class 3300 for 1.050 inches and larger) and shall be of the following types:

a. For waterside of submarine cooler applications constructed for salt water cooling, a full penetration weld with joint construction in conformance with P-70, P-71, or P-72 of MIL-STD-22 shall be used. Nipples shall be composition 70-30 copper-nickel, in accordance with MIL-C-15726.

b. For other applications, in addition to the method specified in a above, the following are acceptable:

(1) Single-J, fillet-reinforced weld V-27 in accordance with MIL-STD-22. Nipples shall be copper-nickel in accordance with MIL-T-16420.

(2) Tapped hole in accordance with SAE J1926/1 fitted with an adapter of composition 70-30 or 90-10 copper-nickel alloy in accordance with MIL-C-15726 (see 3.3.1.1).

c. In addition to methods of b above, for other applications where the connection will not normally be in contact with salt water, a socket weld using a T by 2T minimum fillet in accordance with P-14 of MIL-STD-22 shall be acceptable.

3.1.16.1 High pressure applications. High pressure applications shall lead waterbox vent and drain connections out through a waterbox flange or tube sheet. "High-pressure", for purposes of this specification, refers to waterside pressures above 200 lb/in².

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3.1.17 Mounting. Cooler shall be mounted such that any leakage from the joint between the outer tube sheets and the waterboxes shall not enter the attached equipment. This joint shall be visible from the exterior of the machine without removal of any covers or other machine parts.

3.1.18 Supports. The means of support for air coolers shall be independent of connecting piping. If coolers are supported from a vertical structure, the supports shall be constructed so that bolts in shear shall not constitute the primary means of support. Coolers shall not be supported by plates or brackets in such a manner that the primary means of support is obtained from the bolts securing the frame-tube sheet-waterbox flange joints.

3.1.19 Gasket application. For waterside application of 200 lb/in² design pressure and under, flat gaskets may be used. O-ring gaskets shall be used for design pressures above 200 lb/in². Blind gaskets or packings (those not capable of being replaced without destructive disassembly) shall not be used.

3.1.19.1 Flat gaskets. Flat gaskets shall be 1/16-inch thick. Coolers shall be constructed so that gasket width shall be not less than 5/16 inch. Gasket material shall be in accordance with MIL-PRF-6855, class 1, 80 durometer. Synthetic rubber shall be used, except for applications where design pressure does not exceed 100 lb/in², MIL-PRF-1149, type II, class 5, synthetic rubber may be used.

3.1.19.2 O-ring gaskets. O-ring gaskets shall be in accordance with type I of MIL-G-21610, except for the special applications for adaptors and plugs. In these cases, O-ring gaskets in accordance with class 2 of SAE AMS7259 shall be used. O-ring segments for sealing under waterbox partitions shall be integral with peripheral O-ring. Design retaining grooves for O-ring gaskets shall be in accordance with MIL-STD-2193.

3.1.20 Identification plates. Each cooler shall bear an identification plate in accordance with type A, B, C, D, or L of MIL-DTL-15024 and MIL-P-15024/5 with material choice being limited to the wrought brass, cast brass, cast bronze or corrosion-resisting steel prescribed therein. All plates shall be normal service plate. Provision shall be made for the following information:

a. Name of unit (GENERATOR AIR COOLER, MOTOR AIR COOLER).

b. Class of unit cooler (see 1.2).

c. Manufacturer's service part number.

d. National stock number (allow 20 spaces).

e. Name of manufacturer.

f. Contract or order number. (The contracting activity's contract or purchase order number shall be entered in this space.) (Utilize width of plate to allow maximum number of spaces.)

g. Blank space for Defense Contract Management Agency (DCMA) stamp.

h. Date of manufacture.

i. Serial number.

j. Maximum test pressure, leak-off system.

k. Maximum hydrostatic test pressure, waterside. (For class 1 coolers, no entry shall be made in this space [see 3.1.13].)

l. Four blank spaces for unit number (This space will be used for numbering the stamping to be done by the shipyard, when required for shipboard references purposes).

m. Designation "U.S.".

Design pressure entries shall not appear on identification plates.

3.2 Materials. The materials as specified in 3.2.1 through 3.2.4 shall be used in the construction of the coolers, except that the manufacturer shall have the option of substituting commercial material having equal or better physical and chemical properties than the materials specified herein.

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3.2.1 Class 1 cooler materials. Class 1 coolers shall be constructed of the materials specified in table I. Class 1 coolers have certain structures for which critical dimensions and material analyses shall be required. These are the outer tube sheets; sea water vent and drain connections; inner heat transfer tubes; waterboxes including all parts, subparts, and access covers; and all nuts, studs, bolts and washers which join sea water containing parts (see 4.4.1 and 4.4.2).

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TABLE I. Class 1 cooler materials.

Part	Material	Applicable document
Airside materials (frame, support plates, cover plates, and so forth)	Steel, grade A, class U Steel, grade B, seamless	MIL-S-22698 ASTM A106/A106M
Waterboxes [1/]	Copper-nickel-alloy, composition 70-30 Tubing, copper-nickel-alloy, composition 70-30 Copper-nickel-alloy, composition 70-30; or bronze, nickel aluminum Titanium, grade 2	MIL-C-15726 MIL-T-16420 ASTM B369/B369M or MIL-B-24480 ASTM B265
Outer tube sheets	Copper-nickel-alloy, composition 70-30 Carbon steel plate clad with titanium, grade 2	MIL-C-15726 ASTM A515/A515M, ASTM B265, and ASTM B381
Inner tube sheets	Copper-nickel-alloy Carbon steel plate clad with titanium, grade 2	MIL-C-15726 ASTM A515/A515M, ASTM B265, and ASTM B381
Inner tubes	Copper-nickel-alloy, composition 70-30 Titanium, grade 2	MIL-T-15005 ASTM B338
Outer tubes	Copper Titanium, grade 2	MIL-T-24107 ASTM B338
Fins	Nonferrous	—
Bolts or studs in joints involving sea water tightness or subject to submergence pressure [2/ 3/]	Nickel-copper-aluminum alloy, hot finished, annealed, and age hardened, or cold drawn, annealed, and age hardened	QQ-N-286
Bolts or studs for other service	Nickel-copper, grade 400 or nickel-copper-aluminum, grade 500	MIL-DTL-1222
Nuts for bolts and studs in joints involving sea water tightness or subject to submergence pressure [3/]	Nickel-copper-aluminum alloy Nickel-copper alloy, class A, hot finished Nickel-copper alloy, cold drawn and stress relieved, or hot finished	QQ-N-286 QQ-N-281 MIL-N-24106
Nuts for use on air side	Steel, nylon insert, or nickel-copper alloy Nylon insert	NASM17829 NASM17828

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TABLE I. Class 1 cooler materials – Continued.

Part	Material	Applicable document
Nuts for bolts and studs specified	Nickel-copper, grades 400 and 405	MIL-DTL-1222
Jack screws 3/	Copper-aluminum alloy, copper alloy C61400 Phosphor bronze Copper-silicon alloy	ASTM B171/B171M ASTM B98/B98M ASTM B139/B139M
Zinc protectors	Zinc	MIL-DTL-18001
Vent and drain nipples	Copper-nickel alloy, seamless, composition 70-30	MIL-T-16420
O-ring gaskets 4/	Rubber, synthetic, type I Rubber, synthetic, class 2	MIL-G-21610 SAE AMS7259
Tell-tale system vent and drain connectors 5/	Copper-nickel alloy, composition 70-30	MIL-C-15726

NOTES:

1/ Inspection covers including zinc covers shall be of the same copper alloy as the waterboxes to which fitted.

2/ Bolts and studs shall also be in accordance with MIL-DTL-1222.

3/ Form in accordance with MIL-DTL-1222.

4/ If required, lubricant for O-ring gaskets shall be in accordance with MIL-PRF-24508 or SAE AMS7259.

5/ See 3.3.1.1 and figure 1.

3.2.2 Class 2 cooler materials. Class 2 coolers shall be constructed of any of the materials specified in table I, or those specified in the table II, except that airside materials shall be as specified in table I.

TABLE II. Additional class 2 cooler materials.

Part	Material	Applicable document
Plugs, zinc support 1/	Copper-nickel alloy, composition 70-30	MIL-C-15726
Gaskets 2/	Rubber, synthetic, class 1, grade 80 Rubber, synthetic type II, class 5	MIL-PRF-6855 MIL-PRF-1149

NOTES:

1/ Plugs shall conform to 803-5959186 (see 3.3.5.4).

2/ See 3.1.19.1.

3.2.3 Class 3 cooler materials. Class 3 coolers shall be constructed of the materials specified in table III, except that airside materials shall be as specified in table I. Titanium materials listed in table I may also be used.

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TABLE III. Class 3 cooler materials.

Part	Material	Applicable documents
Waterboxes [1/]	Copper-nickel alloy Tubing, copper-nickel alloy Copper-nickel alloy Bronze, nickel-aluminum or bronze, sand castings, alloy C90300, C90500, or C92200	MIL-C-15726 MIL-T-16420 ASTM B369/B369M MIL-B-24480 or QQ-C-290
Outer and inner tube sheets	Copper-nickel alloy	MIL-C-15726
Inner tubes	Copper-nickel alloy, composition 90-10	MIL-T-15005
Outer tubes	Copper	MIL-T-24107
Fins	Nonferrous	—
Bolts, studs, and nuts	Nickel-copper, N04400, and N04405 or nickel-copper-aluminum, N05500	MIL-DTL-1222
Nuts for use on airside	Steel, nylon insert, or nickel-copper alloy Nylon insert	NASM17829 NASM17828
Jack screws	Copper-aluminum alloy Copper alloy C61400 Phosphor bronze Copper-silicon alloy	QQ-C-465 ASTM B171/B171M ASTM B98/B98M ASTM B139/B139M
Zinc protectors	Zinc	MIL-DTL-18001
Gaskets [2/]	Rubber, synthetic, class 1, grade 80 Rubber, synthetic, type II, class 5	MIL-PRF-6855 MIL-PRF-1149
Plugs, anode support	Copper-nickel alloy, composition 70-30 or 90-10	MIL-C-15726
Washers	Copper-aluminum alloy, copper alloy C61400 Brass, naval	ASTM B171/B171M QQ-B-639
Adaptors	Copper-nickel alloy	MIL-C-15726
O-ring gaskets [3/]	Rubber, synthetic, type I Rubber, synthetic, class 2	MIL-G-21610 SAE AMS7259
Tell-tale system vent and drain connectors [4/]	Copper-nickel alloy	MIL-C-15726

NOTES:

[1/] Inspection covers including zinc covers shall be of the same copper alloy as the waterboxes to which fitted.
[2/] See 3.1.19.1.
[3/] If required, lubricant for O-ring gaskets shall be in accordance with MIL-PRF-24508 or SAE AMS8660.
[4/] See 3.3.1.1 and figure 1.

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3.2.4 Class 4 cooler materials. Class 4 coolers shall be constructed of the materials specified in table IV, except that airside materials shall be as specified in table I. Titanium materials listed in table I may also be used.

TABLE IV. Class 4 cooler materials.

Part	Material	Applicable document
Waterboxes	Copper-nickel alloy	MIL-C-15726
	Tubing, copper-nickel alloy	MIL-T-16420
	Copper-nickel alloy	ASTM B369/B369M
	Bronze, nickel-aluminum	MIL-B-24480
	Bronze, sand castings, alloy C90300, C90500, or C92200	QQ-C-290
Outer and inner tube sheets	Copper-nickel alloy, or nickel-aluminum-bronze, alloy C63000	MIL-C-15726
	Aluminum-bronze, alloy C61400	ASTM B171/B171M
Inner tubes	Copper-nickel alloy, composition 90-10	MIL-T-15005
Outer tubes	Copper	MIL-T-24107
Fins	Nonferrous	—
Nuts, studs, and nuts	Nickel-copper, grades 400 and 405, or nickel-copper-aluminum, grade 500	MIL-DTL-1222
Nuts for use on air side	Steel, nylon insert, or nickel-copper alloy	NASM17829
	Nylon insert	NASM17828
Jack screws	Copper-aluminum alloy, copper alloy C61400	ASTM B171/B171M
	Phosphor bronze	ASTM B98/B98M
	Copper-silicon alloy	ASTM B139/B139M
Gaskets 1/	Rubber, synthetic, class 1, grade 80	MIL-PRF-6855
	Rubber, synthetic, type II, class 5	MIL-PRF-1149
Washers	Copper-aluminum alloy	ASTM B171/B171M
	Copper alloy C61400; or brass, naval	QQ-B-639
Adaptors	Copper-nickel alloy	MIL-C-15726
O-ring gaskets 2/	Rubber, synthetic, type I	MIL-G-21610
	Rubber, synthetic, class 2	SAE AMS7259
Tell-tale system vent and drain connections 3/	Copper-nickel alloy	MIL-C-15726

NOTES:

1/ See 3.1.19.1.

2/ If required, lubricant for O-ring gaskets shall be in accordance with MIL-PRF-24508 or SAE AMS7259.

3/ See 3.3.1.1 and figure 1.

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3.2.5 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Construction.

3.3.1 Double tube and tell-tale provision. Air coolers shall have extended surface outer tubes sealed into inner tube sheets or headers by brazing, welding, or rolling. A tell-tale system shall be provided. These aspects of construction shall be as follows:

a. The outer tubes shall have longitudinal grooves at least 0.018-inch deep in their internal surfaces. No more than 50 percent of the outer tube inner-diameter shall be in contact with the inner tube outer-diameter after the inner tubes are expanded tightly into the outer tubes. After inner tube expansion, there shall be contact between the two tubes to conduct the heat, and the grooves shall readily lead off leakage from any perforation of an inner tube to the end compartments formed between outer tube sheets and inner tubes sheets (headers).

b. Leakage between the two tubes shall be readily detected through the use of a tell-tale system leading from the end compartments with vent and drain connections on each outer tube sheet. These connections shall terminate in straight threaded O-ring connections in accordance with SAE J1926/1 (see 3.3.1.1). These connections shall be located outside the motor or generator so that any leakage past these connections shall not get into shell or air side of the cooler. These connections shall not be plugged except while leak testing the void space.

c. The void space between outer tubes and between inner tube sheet (or header) and outer tube sheet shall be sealed by the tube-to-tube sheet joints as required above and by welding the inner and outer tube sheets. Any hole or other opening which penetrates either tube sheet, other than holes for heat transfer tubes and vents and drains, shall be located outside the weld between tube sheets.

d. Fasteners shall not be welded or brazed to tube sheets.

e. The tell-tale system of the cooler shall withstand an air pressure test of not less than 10 lb/in² with the waterbox removed so that the location of leaking inner tubes or inner tube/outer tubesheet joints can be determined.

f. Hardware that is not normally installed shall not be required to pass the air pressure test.

g. An instruction plate in accordance with MIL-DTL-15024 and MIL-P-15024/5 shall be provided to indicate the function of these holes and to warn against improper plugging of them. This plate shall be located close to one of the leak-off connections.

3.3.1.1 Tell-tale system vents and drains. At each end of coolers having an odd number of waterside passes and at the inlet-outlet end of coolers having an even number of waterside passes, the required vent and drain connections for the tell-tale system shall be made by the method specified in 3.1.16b(2), except that minimum size shall be ¼-inch NPS and shall be at least one NPS smaller than the waterside vent and drain connections (see 3.1.16). O-ring connectors in accordance with figure 1 shall be furnished for each tell-tale system connection.

3.3.2 Tubes. Heat transfer tubes shall be straight and seamless.

3.3.2.1 Tube sizes. Copper-nickel tube sizes shall be as specified below. Titanium tube sizes shall be as necessary for equal or better strength to these.

a. Unless otherwise specified (see 6.2), class 1 coolers shall be provided with inner tubes of ⅝-inch outside diameter (od), 0.065-inch (number 16 Birmingham wire gauge [BWG]) wall thickness.

b. Class 2 and 3 coolers shall be provided with inner tubes of ⅝-inch od, 0.049-inch (number 18 BWG) wall thickness or ½-inch od, 0.049-inch (number 18 BWG) wall thickness.

c. Class 4 coolers shall be provided with either inner tubes of ⅝-inch od, 0.049-inch (number 18 BWG) wall thickness; or ½-inch od, 0.049-inch (number 18 BWG) wall thickness; or ⅜-inch od, 0.035-inch (number 20 BWG) wall thickness.

3.3.2.2 Tube expansion in outer tube sheets. Tubes shall be expanded into the outer tube sheets at both ends by means of a tube expanded in accordance with MIL-E-15809 that shall be governed by an automatic tube expander control. The expanded portion of the tube-to-tube sheet joint shall not approach closer than ⅛ inch to the back face of outer tube sheet. The minimum depth of expansion of tubes shall be as specified in table V.

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TABLE V. Minimum depth of expansion.

Tube od (inch)	Minimum depth of expansion (inch)
5/8	5/8
1/2	5/8
3/8	1/2

3.3.3 Outer tube sheets.

3.3.3.1 Minimum thickness. Tube sheet minimum thickness shall not be less than:

a. For joints with inlet end flare (see 3.3.3.3) – Depth of expansion (see 3.3.2.2), plus depth of flare, plus 1/8 inch.

b. For non-flared joints – Depth of expansion plus 1/8 inch.

3.3.3.2 Tube holes. Holes for tubes in outer tube sheets shall be drilled 0.001 inch larger than the od of the inner tube (after assembly with the outer tube) with a plus tolerance of 0.005 inch.

3.3.3.3 Flaring of holes. The holes for the inlet ends of the tubes shall be flared in accordance with table VI to allow for belling the ends of the tubes.

TABLE VI. Flaring of holes for inlet ends.

Tube od (inch)	Radius of flare (inch)	Diameter of flare, at outside face of tube sheet (inch)
5/8	1/2	3/4
1/2	3/8	5/8
3/8	5/16	1/2

3.3.3.4 Grooving of holes in tube sheets. The holes for tubes in the tube sheets shall be grooved in accordance with table VII.

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TABLE VII. Grooving of holes in tube sheets.

Tube size od	Cooler application	Number of grooves in outer tube sheet 1/	Number of grooves in inner tube sheet 2/	Width of groove (inch)	Depth 3/ of groove (inch)	Space between grooves (inch)
⅝ inch and ½ inch	Class 1 or 2 cooler	3	1	1/16	0.012	⅛
	Class 3 or 4 cooler	2	1	1/16	0.012	⅛
⅜ inch	Class 4 cooler	1	1	1/16	0.012	—

NOTES:

1/ The grooves shall be located in the expanded portion of the joint.

2/ No grooving of inner sheet is required when a welded or brazed joint is used for outer tube-to-inner tube sheet attachment.

3/ Tolerance on groove depth is ±0.005 inch.

3.3.3.5 Rounding of tube hole edges. Where tubes are to be expanded the edges of the holes for the tubes shall be rounded on a 1/16-inch radius except where a belled contour is required for the tube inlet end.

3.3.3.6 Inner tubes inlet ends. The inlet ends of inner tubes shall be expanded and flared (see table VI) and the ends shall then be finished flush with the face of the tube sheet. In no case shall the ends of tubes be below the face of the tube sheets. Discharge ends of tubes shall not protrude more than 3/16-inch beyond the face of the tube sheet.

3.3.4 Support plates. Holes for tubes in support plates shall be not more than 1/64-inch greater (nominal diameter) than the od of the expanded tube. Edges of the holes shall have all roughness removed.

3.3.5 Waterboxes. Waterboxes shall be proportioned to provide sufficient area at all points for flow of the circulating water with the minimum of turbulence and for uniform distribution of the water of all tubes in each pass. Components shall have an inlet waterbox compartment depth, measured normal to the tube sheet, of not less than one-third the diameter of a circle of area equivalent to that of the tube sheet exposed to circulating water flow. Inter-pass (reversing) waterbox compartments shall be deep enough to ensure that the cross section of full flow shall have an area not less than four times the area of the tubes discharging into the compartment. Waterboxes shall be provided with vent holes in the partitions between passes.

3.3.5.1 Waterbox removal. Removal of the waterboxes shall only require removal of the waterbox fasteners. Removal of covers or any other parts of the cooler or parent equipment shall not be required to remove the waterboxes.

3.3.5.2 Water partition plate. Waterbox partition plate width shall be not less than ¼ inch at the gasket contact surface.

3.3.5.3 Jack screws. Jack screws shall be fitted on the waterboxes of coolers of 12-inch and above nominal waterbox diameter.

3.3.5.4 Corrosion preventive anodes. Anodes shall be provided for salt water cooled air coolers using copper-nickel tubes in accordance with MIL-A-19521 with the following exceptions:

a. For class 1 air coolers, threaded support plugs shall not be used. Cover plates secured by threaded fasteners shall be used for support of anodes.

b. For class 2 air coolers, bolted cover plates or threaded plugs shall be used for support of anodes. If used, threaded plugs shall be in accordance with 803-5959186.

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3.3.6 Cyclic life design.

3.3.6.1 Class 1 coolers. For class 1 coolers, the waterside shall withstand cyclic loading as specified (see 6.2) with associated temperature variation in conjunction with piping reaction load below and 150 cycles of the specified hydrostatic test pressure when applied in the waterbox. Diameter and thickness of attached circulating water piping shall be as specified (see 6.2). Piping reaction values shall be as follows:

Piping axial load = $\pi\, r_m\, t \left(s_y - \frac{Pr_i}{2t}\right)$

Piping bending moment = $1.3\, \pi\, {r_m}^2 t \left(s_y - \frac{Pr_i}{2t}\right)$

Piping torsional moment = $\pi\, {r_m}^2 t \sqrt{{s_y}^2 - \left(\frac{Pr_i}{2t}\right)^2}$

Where:

P = design pressure for the component, lb/in²

r_i = pipe inside radius, inches

r_m = pipe mean radius, inches $\left(\frac{r_i + r_o}{2}\right)$

r_o = pipe outside radius, inches

S_y = tabulated value of yield strength of the piping material at the component's design temperature, lb/in²

t = pipe thickness, inches

3.3.6.2 Threaded fasteners. Threaded fasteners securing waterbox inspection openings on class 1 coolers shall be constructed so that 200 inspection cover removal and reinstallation cycles and cyclic loading specified in 3.3.6.1 shall not cause the maximum cumulative usage factor to exceed 0.8 in accordance with section III (class 1 vessel) of ASME BPVC (see 3.1.9.1.6c(5)).

3.3.6.3 Threaded plugs, connections, and closures. Threaded plugs, threaded connections, and threaded closures are prohibited in the submergence pressure boundary of class 1 coolers.

3.3.7 Cyclic life analysis. Except where other analytical methods are permitted herein, boundaries subject to cyclic loading of class 1 coolers shall be analyzed for specified load combinations (for example, internal pressure in combination with that piping reaction load that produces the highest stress) in accordance with the methods and criteria of the latest edition (including revisions, addenda, and applicable cases) of section III of the ASME BPVC, class 1 vessels. Design fatigue curves for materials commonly used in fabrication of coolers and design stress intensity values for materials not covered in section III (class 1 vessels) of ASME BPVC shall be obtained from NAVSEA. Method of analysis for flat, perforated plates shall be in accordance with section III of ASME BPVC. Cumulative usage factor for all transients shall not exceed 0.8.

3.4 Painting. Coolers shall be painted as follows:

a. External and internal nonferrous surfaces shall not be painted.

b. Steel surfaces shall be treated as follows:

(1) After all machining, welding, and brazing operations are completed, all rust and other corrosion products and flux shall be removed by abrasive blasting, sanding, wire brushing, or other mechanical means. This surface shall be thoroughly cleaned of all grease, oil, and dirt by solvent wiping and rinsing, vapor degreasing, or caustic washing followed by rinsing.

(2) Primer shall be applied in accordance with one of the following methods:

(a) One coat of primer pretreatment in accordance with TT-C-490 (0.0002- to 0.0005-inch dry film thickness) followed by primer conforming to TT-P-1757.

(b) A hot-dip tank phosphate treatment conforming to type I of TT-C-490 (0.0006- to 0.001-inch dry film thickness) followed by primer conforming to TT-P-1757.

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(c) Primer conforming to TT-P-1757 shall be applied as one coat (0.0006- to 0.0008-inch dry film thickness).

(3) Two coats of gray enamel conforming to class 2 of MIL-DTL-15090 shall be applied as continuous films approximately 0.001-inch thick each.

3.5 Reliability and maintainability. Each cooler shall have an operating life of 100,000 hours with no scheduled maintenance. Corrective maintenance shall be no more than 2 hours per 20,000 hours of operation, without the use of special tools. Special tools are defined as tools not listed in the GSA Global Supply Catalog.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2).

(b) Conformance inspection (see 4.6).

4.2 First article inspection. First article inspection shall be performed on each size and design of cooler offered when a first article sample is required (see 3.1). This inspection shall include the examinations of 4.5 and the tests specified in table VIII.

4.3 Conformance inspection. Conformance inspection shall include the examinations of 4.5 and the tests specified in table VIII. Every cooler offered to the Government shall be tested in accordance with 4.4.

TABLE VIII. Examinations and tests.

Characteristic	Verification	Requirement	First article inspection	Conformance inspection
Visual and dimensional	4.4	3.1.1 through 3.1.9.1.6, 3.1.14 through 3.1.19.2, 3.1.20, 3.3.2, 3.3.5.4, 3.4	X	X
Materials	4.4	3.2.1 through 3.2.4 as applicable	X	—
Critical dimensions	4.4.1	3.2.1	As required	—
Critical materials	4.4.2	3.2.1	As required	—
Hydrostatic	4.5.1	3.1.11, 3.1.12, 3.1.13	X	X
Leak-off	4.5.1.2	3.3.1	X	X
Performance	4.5.2	—	X	As required
Shock	4.5.3	3.1.10	X	—
Cyclic life	4.6	3.3.6, 3.3.7	X	—
Reliability and maintainability	4.7	3.5	X	—
Prohibited materials	4.8	3.1.5	X	—

4.4 Visual, dimensional, and materials examination. Each cooler offered for delivery shall be examined for structural, identification plates, construction, painting, materials, and any other requirements specified herein not involving tests. Both the assembled tube bundle (including tube internals) and the shell shall be examined to the requirements listed in table VIII. Class 1 coolers shall be subject to the additional detailed examinations of 4.4.1 and 4.4.2.

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4.4.1 Critical dimensions. For class 1 coolers, critical dimensions such as waterbox thicknesses, tube sheet thicknesses, tube hole sizes, and tube sheet ligaments shall be measured. A critical dimension is defined as a dimension based on fatigue analysis or directly related to another dimension so based (for example, the size of a tube hole). For class 1 coolers, a 100-percent examination of the tube sheet holes and ligament widths shall be made.

4.4.2 Critical materials. For class 1 coolers, the following items are designated as critical materials. Each of these items shall have material traceability including heat lots, chemical and physical properties. Critical materials consist of the following:

a. Outer tube sheets.

b. Sea water vent and drain connections.

c. Heat transfer tubes (inner tubes only).

d. Waterboxes including all parts, subparts, and access covers.

e. Nuts, studs, bolts, and washers, joining sea water containing parts.

4.5 Tests. Except for the high impact shock tests, those tests that require assembly with the parent machine for which the cooler is constructed shall be conducted at the plant of the parent machine manufacturer.

4.5.1 Pressure tests. Each air cooler shall be given the following waterside and leak-off system hydrostatic pressure tests:

4.5.1.1. Waterside hydrostatic. Each cooler offered for delivery shall be given a separate hydrostatic test of the shell side, tube side, and void spaces (between double tubesheets), as applicable. Coolers shall be tested at 1½ times design pressure. Double-tubesheet void spaces and leakage warning passages shall be tested to the same pressure as the coolant side. Unless otherwise specified (see 6.2), test pressures shall be held for not less than 30 minutes, and leakage shall not occur.

4.5.1.2. Leak-off system. Each cooler offered for delivery shall be given a pressure test using air, at a minimum air pressure of 10 lb/in^2 (minimum). Acceptance criteria shall be no leakage.

4.5.2 Performance tests installed on parent machine. Where tests are required by the specifications for the parent machine on a selective or periodic basis, these shall be conducted with the air cooler installed. When performance tests of the parent machine are required, the cooler shall be supplied with cooling water at rated flow and temperature. Test measurements shall include the following:

a. Water flow rate (gal/min).

b. Water inlet temperature (°F).

c. Water outlet temperature (°F).

d. Water pressure drop across cooler (lb/in^2).

e. Air inlet temperature (°F).

f. Air outlet temperature (°F).

g. Air pressure drop across cooler (inches H_2O).

4.5.2.1 Calculated values. From the testing of 4.5.2, the following values shall be calculated:

a. When the parent equipment is being shock tested, the air cooler shall be shock tested in its installed position.

b. The waterside of the air cooler shall be filled with fresh water and pressurized to design operating pressure during the test. All equipment or parts shall be mounted on the shock machine.

c. Heat transfer rate (clean tube condition) (Btu per hour per square foot outside surface per °F logarithmic mean temperature difference).

4.5.3 High impact shock test. The test shall be as specified for grade A, class 1 in MIL-DTL-901 and as follows:

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a. When the parent equipment is being shock tested, the air cooler shall be shock tested in its installed position.

b. The waterside of the air cooler shall be filled with fresh water and pressurized to design operating pressure during the test. All equipment or parts shall be mounted on the shock machine in a manner simulating the actual installation on board ship. The weight designated of the shock test shall be as required by the combined weight of the equipment being tested. When the cooler is being tested independent of the parent equipment, grade 2 bolts in accordance with MIL-DTL-1222 shall be used to hold down the cooler during the shock test.

c. Failure to perform principal function. Any of the following failures shall be considered as cause for rejection of the equipment:

(1) Breaking of any parts.

(2) Appreciable distortion or dislocation of any parts such as mounting feet, tubes, baffles, or attached piping.

(3) Leakage of water from the cooling waterside of the cooler.

(4) Following the shock test, the leak-off side of the cooler shall be pressurized with air to design test pressure. There shall be no pressure drop indicating leakage in the leak-off portion of the cooler.

d. Disposal of shock-tested equipment.

(1) Equipment which has been subjected to the high-impact shock test and has failed to perform any of the principal functions specified herein shall not be acceptable, either in whole or any of the parts.

(2) Equipment which has been subjected to the high-impact, shock test and has successfully performed all the principal functions specified herein will not be considered for acceptance until it is first disassembled, inspected, and refurbished.

e Shock test extension. Unless otherwise specified (see 6.2), extension of shock testing shall not be permitted. An air cooler of each size and similar construction shall be shock tested.

4.6 Cyclic loading proof test – special case (cyclic loading stipulated, class 1 coolers). For the special case of an inlet-outlet waterbox not amenable to analysis for determination of adequacy for cyclic life requirements, the adequacy of the waterbox or a scale model thereof shall be verified by a proof test. This proof test shall consist of a qualitative survey of the waterbox utilizing brittle coating (stress coat or its equivalent), or photostress (molded birefringent coatings) techniques as deemed suitable, in order to determine the location of maximum strain. Quantitative measurement of the magnitude of maximum principal strain in the waterbox shall be obtained from strain gauges of ⅛-inch gauge length or smaller. Waterbox loadings shall be comprised of a combination of design submergence pressure in conjunction with piping reaction load (that load which will produce highest stress) applied to the nozzles. The nozzle loads shall be calculated as specified in 3.3.6.1. A loading jig utilizing mechanical or hydraulic jacks shall be provided to apply external loads to the nozzle flanges. The following combinations of loads shall be applied:

a. Internal pressure and external piping load (that piping reaction load which produces the highest stress) applied to one flange.

b. Internal pressure and equal external loads (that piping reaction load which provided the highest stress) applied to both flanges.

Since the angular orientations of the applied bending moments to give the maximum stress for each load combination are unknown, and the location of maximum stress is also unknown, the angular orientations of the applied bending moments for load combinations (a) and (b) shall correspond to increments of 45 degrees around the nozzles from 0 to 360 degrees in an initial run, and a minimum of 9-degree increments shall be used within the 45-degree maximum stress sector in a follow-on run. The location, magnitude, and direction of the maximum principal strain (stress) can be determined by brittle coating or photostress, but must be checked with electric resistance strain gauges. If the maximum range of strain (stress) intensities thus determined in any location of the waterbox (by either load combination (a) or (b) does not exceed the allowed strain (stress) intensity range for the stipulated cycle life as determined by Fatigue Curve for the particular material involved, the waterbox design shall be considered acceptable.

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4.7 Reliability and maintainability tests. The reliability of the cooler shall be demonstrated by the successful completion of all operational tests (see 4.5) and reliability predictions. The prediction guidance provided in MIL-HDBK-217 may be used. The maintenance procedures, removal of parts for repair, and the time required for accomplishment shall be verified during first article testing. The demonstration procedures in MIL-HDBK-470 may be used.

4.8 Prohibited materials. Prohibited materials shall be verified as specified (see 3.1.5 and 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The air coolers covered by this specification are intended for heat transfer equipment used to cool the air circulating within electrical equipment such as motors and generators installed on naval ships. Current stock of air coolers may be used until depleted; future requisitions for air coolers these should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class of cooler (see 1.2).

c. When first article is required (see 3.1).

d. The waterside design pressure, lb/in² (see 3.1.13).

e. For submarine air coolers, dimensions of hatch, bulkhead opening, or other access openings available for removal of cooler from the compartment in which the unit will be installed (see 3.1.2.1).

f. Ship motion and attitude requirements, if other than specified (see 3.1.3).

g. Requirement for prohibited materials verification (see 3.1.5 and 4.8).

h. Maximum permissible pressure drop across the waterside (lb/in²), if other than specified (see 3.1.11).

i. Inlet temperature of water (°F), if other than specified (see 3.1.12).

j. Heat transfer duty, Btu/hr (see 3.1.14).

k. Air outlet temperature (°F), if other than specified (see 3.1.14).

l. Requirements for inlet and outlet water connections, if other than specified (see 3.1.15).

m. For class 1 coolers, the diameter and thickness of inner tubes, if other than specified (see 3.3.2.1).

n. For class 1 coolers, cyclic life requirements (see 3.3.6.1).

o. For class 1 coolers, the diameter and thickness of attached circulating water piping (see 3.3.6.1).

p. Test pressure requirements, if other than specified (see 4.5.1).

q. Rate of water flow, if established (gal/min) (see 4.5.2).

r. Requirements for the extension of shock testing, if other than specified (see 4.5.3.e).

s. Packaging requirements (see 5.1).

t. Provisioning requirements (see 6.3).

u. Recommended documentation (see 6.5).

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6.3 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified (see 6.2).

6.4 First article inspection. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, samples selected from the first lot size of production items, a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.2. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Recommended documentation. The following list of documentation is recommended to be requested as part of the acquisition of any valve under this specification (see 6.2):

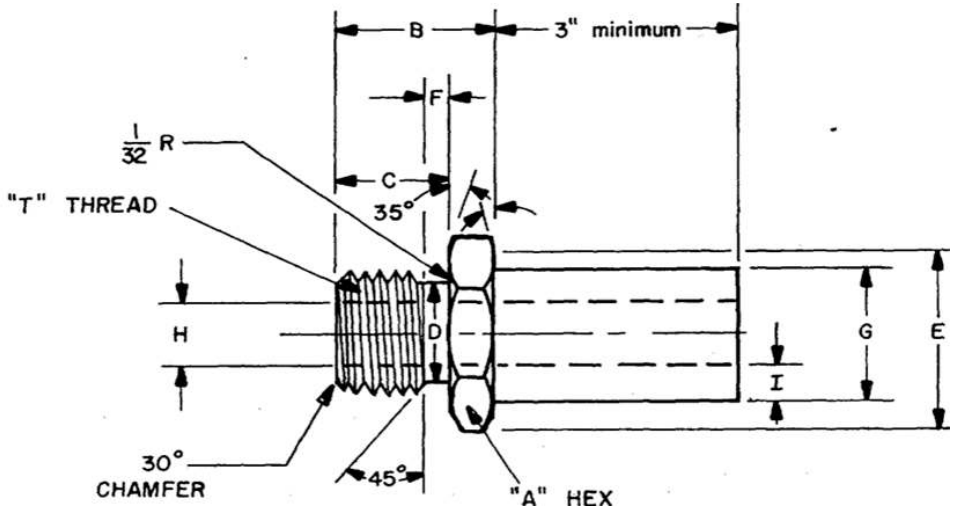
a. Engineering drawings.
b. Materials listing.
c. Installation drawings.
d. Technical manual.
e. Test reports.
f. Certifications.
g. Onboard repair parts list.
h. Design report.
i. Aperture cards.

6.6 Subject term (key word) listing.

Double tube
Electrical equipment
Enclosure
Heat transfer
Tell-tale
Waterbox

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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NOTE: The plug portion of this O-ring connector conforms dimensionally to MS18229.

Dimensions of O-ring connector (all dimensions in inches).

Nominal size	"T" straight thread (UN-2A)	A hexagon across flats	B ±1/64	C ±0.005	D +0.002 -0.003	E ±1/64	F ±0.005	G ±0.005	H	I Minimum	O-ring SAE AS58
1/4	7/16-20	9/16		0.360	0.364	5/8	0.093	0.540	0.203 ±0.003	0.120	-904
3/8	9/16-18	11/16	11/16	0.391	0.482	3/4	0.097	0.675	0.281 ±0.004	0.148	-906
1/2	3/4-16	7/8	3/4	0.438	0.660	15/16	0.100	0.840	0.422 ±0.004	0.203	-908
3/4	1 1/16-12	1 1/4	1	0.594	0.945	1 5/16	0.130	1.050	0.656 ±0.005	0.148	-912
L	1 5/16-12	1 1/2	1	0.594	1.195	1 9/16	0.130	1.315	0.875 +0.007 -0.005	0.180	-916

FIGURE 1. O-ring connector for vents and drains

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CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 4420-2020-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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